

How do we sketch the phase portrait of a node?

Answer:

Sketching a phase portrait for a node involves a systematic process that relies entirely on the eigenvalues and eigenvectors of the system's underlying matrix. A node is an equilibrium point where all trajectories either move directly towards it (stable node) or directly away from it (unstable node).

This behavior occurs when the matrix $\mathbf{A}$ of the system $\mathbf{x}' = \mathbf{Ax}$ has **two real eigenvalues of the same sign**.

- If both eigenvalues are **negative**, it's a **stable node (a sink)**. All solutions approach the origin as $t \rightarrow \infty$.
- If both eigenvalues are **positive**, it's an **unstable node (a source)**. All solutions move away from the origin as $t \rightarrow \infty$.

Step-by-Step Guide to Sketching a Node

Consider the 2x2 SDE: $\mathbf{x}' = \mathbf{Ax}$

Step 1: Find Eigenvalues

First, find the eigenvalues of the matrix $\mathbf{A}$.

You will get two eigenvalues, λ_1 and λ_2 . For the equilibrium at the origin to be a node, λ_1 and λ_2 must be **real numbers** and have the **same sign**.

- If $\lambda_1, \lambda_2 < 0$, you have a **stable node**.
- If $\lambda_1, \lambda_2 > 0$, you have an **unstable node**.

Step 2: Find the Eigenvectors

For each eigenvalue, find its corresponding eigenvector. You will get two eigenvectors, $\mathbf{v}_1$ and $\mathbf{v}_2$, corresponding to λ_1 and λ_2 .

These eigenvectors are crucial because they represent **straight-line trajectories** in the phase portrait.

Step 3: Sketch the Eigenvector Lines

Draw the equilibrium point, which is always the origin $(0, 0)$. Then, draw two lines that pass through the origin in the direction of your two eigenvectors, $\mathbf{v}_1$ and $\mathbf{v}_2$. These lines are the "guiding axes" for your entire sketch.

Step 4: Determine the Direction of Flow on the Eigenvectors

Look at the sign of your eigenvalues.

- **Stable Node ($\lambda < 0$):** The flow is towards the origin. Draw arrows on both eigenvector lines pointing **inward** towards (0, 0).
- **Unstable Node ($\lambda > 0$):** The flow is away from the origin. Draw arrows on both eigenvector lines pointing **outward** from (0, 0).

Step 5: Sketch the Other Trajectories

The shape of the other (curved) trajectories depends on the **magnitudes** of the eigenvalues.

Let's assume $\lambda_1 < \lambda_2$.

- **Behavior Near the Origin:** As trajectories get very close to the origin, they become almost parallel to the eigenvector associated with the **eigenvalue of smaller magnitude** (λ_1 in this case). The "slow" direction dominates near the equilibrium.
- **Behavior Far from the Origin:** As trajectories are far from the origin, they are almost parallel to the eigenvector associated with the **eigenvalue of larger magnitude** (λ_2 in this case). The "fast" direction dominates far from the equilibrium.

For a Stable Node ($\lambda_1 < \lambda_2 < 0$): All trajectories approach the origin. The starting "tail" of the trajectories is far away nearly parallel to the $\mathbf{v}_1$ line (the "smaller" eigenvector) and then curve to approach the origin with the ending "tail" nearly parallel to the $\mathbf{v}_2$ line (the "larger" eigenvector).

For an Unstable Node ($0 < \lambda_1 < \lambda_2$): All trajectories move away from the origin. The starting "tail" of the trajectories are near the origin nearly parallel to the $\mathbf{v}_1$ line (the "smaller" eigenvector) and then curve to move away with the ending "tail" nearly parallel to the $\mathbf{v}_2$ line (the "larger" eigenvector).

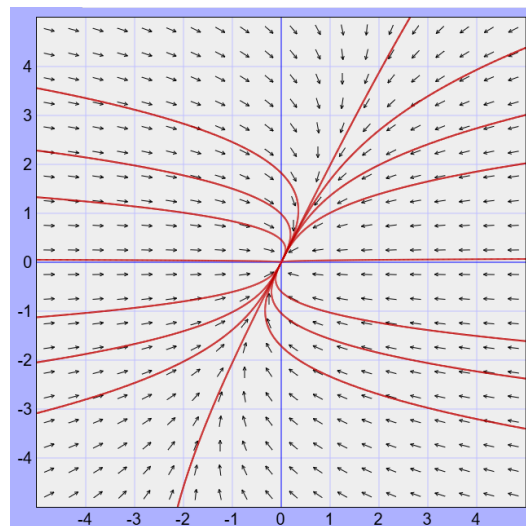
Scenario 1: Sketching a Stable Node

Let's sketch the phase portrait for the system: $\mathbf{x}' = \begin{pmatrix} -3 & 1 \\ 0 & -1 \end{pmatrix} \mathbf{x}$

1. **Eigenvalues:** Since the matrix is upper triangular, the eigenvalues are the diagonal entries: $\lambda_1 = -1$ and $\lambda_2 = -3$. They are both real and negative, so we have a **stable node**.
2. **Eigenvectors:**
 - For $\lambda_1 = -1$: The eigenvector is $\mathbf{v}_1 = (1, 2)$.
 - For $\lambda_2 = -3$: The eigenvector is $\mathbf{v}_2 = (1, 0)$.

3. **Sketch Eigenvector Lines:** Draw a line through the origin along the vector $(1, 2)$ and another line along $(1, 0)$ (the x-axis).
4. **Direction of Flow:** Since both eigenvalues are negative, draw arrows on both lines pointing towards the origin.
5. **Sketch Other Trajectories:**
 - We have $\lambda_1 = -1$ and $\lambda_2 = -3$.
 - The smaller eigenvalue is $\lambda_2 = -3$, associated with $\mathbf{v}_2 = (1, 0)$.
 - Therefore, all trajectories will start far away being more parallel to the $\mathbf{v}_2$ line (the x-axis).
 - They will approach the origin tangent (parallel) to the $\mathbf{v}_1$ line.

The sketch would look like this:



Scenario 2: Sketching an unstable Node

Let's sketch the phase portrait for the system: $\mathbf{x}' = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix} \mathbf{x}$

1. **Eigenvalues:** Since the matrix is upper triangular, the eigenvalues are the diagonal entries: $\lambda_1 = 3$ and $\lambda_2 = 1$. They are both real and positive, so we have an **unstable node**.
2. **Eigenvectors:**
 - For $\lambda_1 = 3$: The eigenvector is $\mathbf{v}_1 = (1, 0)$.
 - For $\lambda_2 = 1$: The eigenvector is $\mathbf{v}_2 = (1, -1)$.

3. **Sketch Eigenvector Lines:** Draw a line through the origin along the vector $(1, 0)$ and another line along $(1, -1)$.
4. **Direction of Flow:** Since both eigenvalues are positive, draw arrows on both lines pointing away from the origin.
5. **Sketch Other Trajectories:**
 - We have $\lambda_1 = 3$ and $\lambda_2 = 1$.
 - The smaller eigenvalue is $\lambda_2 = 1$, associated with $\mathbf{v}_2 = (1, -1)$.
 - Therefore, all trajectories will start near the origin tangent to the $\mathbf{v}_2$ line.
 - They will move away and become more parallel to the $\mathbf{v}_1$ line.

The sketch would look like this:

